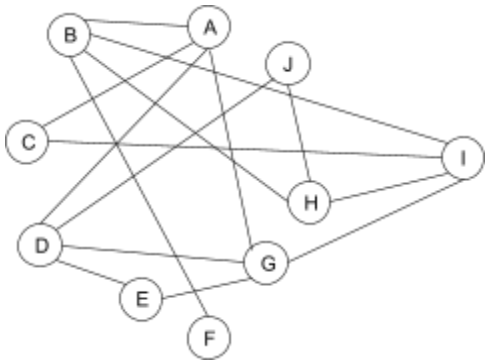
American Computer Science League

2024-2025 ● Contest 4: Short Problem Solutions ● Elementary Division

1. Graph Theory

If you draw the graph, there are at least 2 edges that go out of each vertex except for F which is Fran.



C. Fran

2. Graph Theory

Using inspection, there are 3 ways to get from city B to city C. They are BAC, BADC, and BADEFC.

B. 3

3. Graph Theory

To connect the overlapping networks of NPR and QST, there are 9 possible edges that could be used: P could connect to Q, S, or T; N could connect to Q, S, or T; and R could connect to Q, S, or T.

C. 9

4. Graph Theory

Any traversal that has exactly 7 edges starting with V and ending with X or vice versa is OK because they both have a degree of 3. There is only 1 with 7 edges that all connect correctly: VUXYZVWX.

D. VUXYZVWX

5. Graph Theory By inspection they are: 4 with only 3 vertices - ADCA and ACDA, BEFB and BFEB; 2 with only 4 vertices - BCDFB and BFDCB; 4 with 5 vertices - ACBFDA and ADFBCA, BCDFEB and BEFDCB; and 2 with all 6 vertices - ADFEBCA and ACBEFDA. That makes 12 total cycles.	E. 12
6. Graph Theory Graph #1 has 2 odd vertices (B and C), graph #2 also has 2 odd vertices (B and D), and graph #3 has 0 odd vertices so all 3 are traversable.	D. 1, 2, 3